Active Maintenance Report: aronow_etal_2015

Aronow, Coppock, Crawford, and Green (2015, JSSAM) — Combining List Experiment and Direct Question Estimates

2026-03-18

1 Paper overview

Citation: Aronow, P.M., Coppock, A., Crawford, F.W., and Green, D.P. (2015). Combining List Experiment and Direct Question Estimates of Sensitive Behavior Prevalence. *Journal of Survey Statistics and Methodology*, 3(1), 43–66.

Replication archive: https://doi.org/10.7910/DVN/29984

Summary: The paper proposes a nonparametric combined estimator of sensitive-behavior prevalence that incorporates both a list experiment and a direct question. Under a Monotonicity assumption (no false confessions), direct question admitters reveal their true status, while list experiments identify withholder prevalence. The combined estimator is shown to be asymptotically more efficient than the conventional difference-in-means list estimator. Two placebo tests are derived: Placebo Test I checks whether the list estimate among direct admitters equals 1 (a joint test of No Liars, No Design Effects, and Monotonicity); Placebo Test II tests treatment independence of direct question responses. Both estimator and tests are applied to an original two-study (n = 1,014) Amazon Mechanical Turk experiment and a preregistered replication (n = 1,012) across five sensitive items.

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2 Original archive reproducibility

Table 1: Original archive reproducibility (Category 1 = no errors; 2 = resolvable errors)

Script	Status on Current R	Resolution	Category

ACCG_list_main_experiment.R	setwd() error	Comment out setwd(); run from extracted/ with programs/ and data/ subdirectories	2
ACCG_list_replication_experiment.R	setwd() error; missing library(xtable)	Same setwd fix; add library(xtable) alongside library(lmtest)	2
ACCG_list_placebo_I_power.R	setwd() error; three code bugs	setwd fix; rename listsimulator -> list_simulator; dot -> underscore arg names; $p_value -> [["p_value"]]	2
ACCG_list_placebo_II_power.R	setwd() error	Comment out setwd()	2

Classification: Category 2 (resolvable). All four scripts fail due to a hardcoded `setwd("/Users/Alex/...")` that does not exist in the current environment. Additionally, the replication script omits `library(xtable)`, and the Placebo Test I power script contains three code bugs (described in Section 4). After patching, all scripts run cleanly against the current R environment.

Archive structure note: The original zip places all files at the root level, but the scripts expect `programs/` and `data/` subdirectories. Running from an `extracted/` directory with those subdirectories created resolves the path issues.

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3 Number-by-number comparison

Table 2: Ground truth comparison: key numerical claims (160 rows; selected shown)

Location	Quantity	Paper	Script	Match
table_3	Study A Nuclear power direct est	0.656	0.656	yes
table_3	Study A Nuclear power direct SE	0.021	0.021	yes
table_3	Study A Nuclear power conv est	0.748	0.748	yes
table_3	Study A Nuclear power conv SE	0.084	0.084	yes
table_3	Study A Nuclear power combined est	0.666	0.666	yes
table_3	Study A Nuclear power combined SE	0.048	0.048	yes
table_3	Study A Nuclear power var reduction	67.100	67.100	yes
table_3	Study A Public transportation direct est	0.538	0.538	yes
table_3	Study A Public transportation direct SE	0.022	0.022	yes
table_3	Study A Public transportation conv est	0.513	0.513	yes
table_3	Study A Public transportation conv SE	0.072	0.072	yes
table_3	Study A Public transportation combined est	0.627	0.627	yes
table_3	Study A Public transportation combined SE	0.049	0.049	yes
table_3	Study A Public transportation var reduction	54.000	54.000	yes
table_3	Study A Spanish speaking direct est	0.102	0.102	yes
table_3	Study A Spanish speaking direct SE	0.014	0.014	yes
table_3	Study A Spanish speaking conv est	0.035	0.035	yes
table_3	Study A Spanish speaking conv SE	0.079	0.079	yes
table_3	Study A Spanish speaking combined est	0.042	0.042	yes
table_3	Study A Spanish speaking combined SE	0.074	0.074	yes
table_3	Study A Spanish speaking var reduction	14.000	14.000	yes
table_3	Study A Muslim teachers direct est	0.110	0.110	yes

table_3	Study A Muslim teachers direct SE	0.014	0.014	yes
table_3	Study A Muslim teachers conv est	0.166	0.166	yes
table_3	Study A Muslim teachers conv SE	0.081	0.081	yes
table_3	Study A Muslim teachers combined est	0.187	0.187	yes
table_3	Study A Muslim teachers combined SE	0.074	0.074	yes
table_3	Study A Muslim teachers var reduction	15.300	15.300	yes
table_3	Study A CNN direct est	0.444	0.444	yes
table_3	Study A CNN direct SE	0.022	0.022	yes
table_3	Study A CNN conv est	0.338	0.338	yes
table_3	Study A CNN conv SE	0.105	0.105	yes
table_3	Study A CNN combined est	0.533	0.533	yes
table_3	Study A CNN combined SE	0.070	0.070	yes
table_3	Study A CNN var reduction	54.900	54.900	yes
table_4	Study A Placebo I Nuclear power beta	1.054	1.054	yes
table_4	Study A Placebo I Nuclear power SE	0.095	0.095	yes
table_4	Study A Placebo I Nuclear power p-value	0.568	0.568	yes
table_4	Study A Placebo I Nuclear power n	328.000	328.000	yes
table_4	Study A Placebo I Public transportation beta	0.790	0.790	yes
table_4	Study A Placebo I Public transportation SE	0.091	0.091	yes
table_4	Study A Placebo I Public transportation p-value	0.021	0.021	yes
table_4	Study A Placebo I Public transportation n	269.000	269.000	yes
table_4	Study A Placebo I Spanish speaking beta	0.848	0.848	yes
table_4	Study A Placebo I Spanish speaking SE	0.279	0.279	yes
table_4	Study A Placebo I Spanish speaking p-value	0.585	0.585	yes
table_4	Study A Placebo I Spanish speaking n	51.000	51.000	yes
table_4	Study A Placebo I Muslim teachers beta	1.008	1.008	yes
table_4	Study A Placebo I Muslim teachers SE	0.237	0.237	yes
table_4	Study A Placebo I Muslim teachers p-value	0.973	0.973	yes
table_4	Study A Placebo I Muslim teachers n	55.000	55.000	yes
table_4	Study A Placebo I CNN beta	0.696	0.696	yes
table_4	Study A Placebo I CNN SE	0.143	0.143	yes
table_4	Study A Placebo I CNN p-value	0.034	0.034	yes
table_4	Study A Placebo I CNN n	222.000	222.000	yes
table_5	Study A Placebo II Nuclear power beta	0.066	0.066	yes
table_5	Study A Placebo II Nuclear power SE	0.042	0.042	yes
table_5	Study A Placebo II Nuclear power p-value	0.118	0.118	yes
table_5	Study A Placebo II Public transportation beta	0.000	0.000	yes
table_5	Study A Placebo II Public transportation SE	0.045	0.045	yes
table_5	Study A Placebo II Public transportation p-value	0.994	0.994	yes
table_5	Study A Placebo II CNN beta	-0.056	-0.056	yes
table_5	Study A Placebo II CNN SE	0.045	0.045	yes
table_5	Study A Placebo II CNN p-value	0.206	0.206	yes
table_6	Study B Nuclear power direct est	0.603	0.603	yes
table_6	Study B Nuclear power direct SE	0.022	0.022	yes
table_6	Study B Nuclear power conv est	0.499	0.499	yes
table_6	Study B Nuclear power conv SE	0.089	0.089	yes
table_6	Study B Nuclear power combined est	0.624	0.624	yes
table_6	Study B Nuclear power combined SE	0.052	0.052	yes
table_6	Study B Nuclear power var reduction	66.000	66.000	yes
table_6	Study B Public transportation direct est	0.539	0.539	yes
table_6	Study B Public transportation direct SE	0.022	0.022	yes
table_6	Study B Public transportation conv est	0.578	0.578	yes
table_6	Study B Public transportation conv SE	0.073	0.073	yes
table_6	Study B Public transportation combined est	0.608	0.608	yes
table_6	Study B Public transportation combined SE	0.051	0.051	yes
table_6	Study B Public transportation var reduction	51.900	51.900	yes
table_6	Study B Spanish speaking direct est	0.113	0.113	yes

table_6	Study B Spanish speaking direct SE	0.014	0.014	yes
table_6	Study B Spanish speaking conv est	0.144	0.144	yes
table_6	Study B Spanish speaking conv SE	0.083	0.083	yes
table_6	Study B Spanish speaking combined est	0.149	0.149	yes
table_6	Study B Spanish speaking combined SE	0.076	0.076	yes
table_6	Study B Spanish speaking var reduction	16.400	16.400	yes
table_6	Study B Muslim teachers direct est	0.103	0.103	yes
table_6	Study B Muslim teachers direct SE	0.013	0.013	yes
table_6	Study B Muslim teachers conv est	0.107	0.107	yes
table_6	Study B Muslim teachers conv SE	0.083	0.083	yes
table_6	Study B Muslim teachers combined est	0.123	0.123	yes
table_6	Study B Muslim teachers combined SE	0.074	0.074	yes
table_6	Study B Muslim teachers var reduction	20.100	20.100	yes
table_6	Study B CNN direct est	0.496	0.496	yes
table_6	Study B CNN direct SE	0.022	0.022	yes
table_6	Study B CNN conv est	0.645	0.645	yes
table_6	Study B CNN conv SE	0.101	0.101	yes
table_6	Study B CNN combined est	0.587	0.587	yes
table_6	Study B CNN combined SE	0.064	0.064	yes
table_6	Study B CNN var reduction	60.200	60.200	yes
table_7	Study B Placebo I Nuclear power beta	0.881	0.881	yes
table_7	Study B Placebo I Nuclear power SE	0.113	0.113	yes
table_7	Study B Placebo I Nuclear power p-value	0.294	0.294	yes
table_7	Study B Placebo I Nuclear power n	310.000	310.000	yes
table_7	Study B Placebo I Public transportation beta	0.913	0.913	yes
table_7	Study B Placebo I Public transportation SE	0.091	0.091	yes
table_7	Study B Placebo I Public transportation p-value	0.339	0.339	yes
table_7	Study B Placebo I Public transportation n	277.000	277.000	yes
table_7	Study B Placebo I Spanish speaking beta	0.767	0.767	yes
table_7	Study B Placebo I Spanish speaking SE	0.229	0.229	yes
table_7	Study B Placebo I Spanish speaking p-value	0.309	0.309	yes
table_7	Study B Placebo I Spanish speaking n	58.000	58.000	yes
table_7	Study B Placebo I Muslim teachers beta	0.700	0.700	yes
table_7	Study B Placebo I Muslim teachers SE	0.285	0.285	yes
table_7	Study B Placebo I Muslim teachers p-value	0.293	0.293	yes
table_7	Study B Placebo I Muslim teachers n	53.000	53.000	yes
table_7	Study B Placebo I CNN beta	0.847	0.847	yes
table_7	Study B Placebo I CNN SE	0.135	0.135	yes
table_7	Study B Placebo I CNN p-value	0.256	0.256	yes
table_7	Study B Placebo I CNN n	255.000	255.000	yes
table_8	Study B Placebo II Nuclear power beta	-0.086	-0.086	yes
table_8	Study B Placebo II Nuclear power SE	0.043	0.043	yes
table_8	Study B Placebo II Nuclear power p-value	0.045	0.045	yes
table_8	Study B Placebo II Public transportation beta	0.034	0.034	yes
table_8	Study B Placebo II Public transportation SE	0.044	0.044	yes
table_8	Study B Placebo II Public transportation p-value	0.434	0.434	yes
table_8	Study B Placebo II CNN beta	0.132	0.132	yes
table_8	Study B Placebo II CNN SE	0.044	0.044	yes
table_8	Study B Placebo II CNN p-value	0.003	0.003	yes
table_a1	Factorial Nuclear power direct diff	0.053	0.053	yes
table_a1	Factorial Nuclear power direct SE	0.030	0.030	yes
table_a1	Factorial Nuclear power conv diff	0.249	0.249	yes
table_a1	Factorial Nuclear power conv SE	0.122	0.122	yes
table_a1	Factorial Nuclear power combined diff	0.042	0.042	yes
table_a1	Factorial Nuclear power combined SE	0.071	0.071	yes
table_a1	Factorial CNN direct diff	-0.052	-0.052	yes
table_a1	Factorial CNN direct SE	0.031	0.031	yes

table_a1	Factorial CNN conv diff	-0.307	-0.307	yes
table_a1	Factorial CNN conv SE	0.146	0.146	yes
table_a1	Factorial CNN combined diff	-0.054	-0.054	yes
table_a1	Factorial CNN combined SE	0.095	0.095	yes
table_a2	Replication Study A Nuclear power direct est	0.646	0.646	yes
table_a2	Replication Study A Nuclear power conv est	0.575	0.575	yes
table_a2	Replication Study A Nuclear power combined est	0.647	0.647	yes
table_a2	Replication Study A Nuclear power var reduction	67.556	67.600	yes
table_a2	Replication Study A CNN direct est	0.407	0.407	yes
table_a2	Replication Study A CNN conv est	0.256	0.256	yes
table_a2	Replication Study A CNN combined est	0.298	0.298	yes
table_a2	Replication Study A CNN var reduction	45.350	45.300	yes
table_a3	Replication Study A Placebo I CNN beta	0.711	0.711	yes
table_a3	Replication Study A Placebo I CNN SE	0.133	0.133	yes
table_a3	Replication Study A Placebo I CNN p-value	0.029	0.029	yes
table_a3	Replication Study A Placebo I CNN n	206.000	206.000	yes
table_a5	Replication Study B Nuclear power direct est	0.591	0.591	yes
table_a5	Replication Study B Nuclear power conv est	0.364	0.364	yes
table_a5	Replication Study B Nuclear power combined est	0.521	0.521	yes
table_a5	Replication Study B Nuclear power var reduction	61.053	61.100	yes
table_a5	Replication Study B CNN direct est	0.457	0.457	yes
table_a5	Replication Study B CNN conv est	0.586	0.586	yes
table_a5	Replication Study B CNN combined est	0.604	0.604	yes
table_a5	Replication Study B CNN var reduction	51.321	51.300	yes
table_a6	Replication Study B Placebo I Nuclear power beta	0.739	0.739	yes
table_a6	Replication Study B Placebo I Nuclear power SE	0.110	0.110	yes
table_a6	Replication Study B Placebo I Nuclear power p-value	0.018	0.018	yes
table_a6	Replication Study B Placebo I Nuclear power n	299.000	299.000	yes
table_a6	Replication Study B Placebo I Public transportation beta	0.795	0.795	yes
table_a6	Replication Study B Placebo I Public transportation SE	0.097	0.097	yes
table_a6	Replication Study B Placebo I Public transportation p-value	0.034	0.034	yes
table_a6	Replication Study B Placebo I Public transportation n	263.000	263.000	yes
table_a6	Replication Study B Placebo I Spanish speaking beta	0.301	0.301	yes
table_a6	Replication Study B Placebo I Spanish speaking SE	0.262	0.262	yes
table_a6	Replication Study B Placebo I Spanish speaking p-value	0.008	0.008	yes
table_a6	Replication Study B Placebo I Spanish speaking n	50.000	50.000	yes
table_a7	Replication Study B Placebo II CNN beta	0.092	0.092	yes
table_a7	Replication Study B Placebo II CNN SE	0.044	0.044	yes
table_a7	Replication Study B Placebo II CNN p-value	0.038	0.038	yes
table_a8	Replication Factorial CNN conv diff	-0.330	-0.330	yes
table_a8	Replication Factorial CNN conv SE	0.146	0.146	yes
table_a8	Replication Factorial CNN combined diff	-0.306	-0.306	yes
table_a8	Replication Factorial CNN combined SE	0.105	0.105	yes

All 160 numerical claims match the published paper to reported precision. No discrepancies found. The appendix tables (A2–A8) report variance reductions to three decimal places (e.g., 67.556 for replication Study A nuclear power); the main tables report to one decimal place (67.1). The maintained rewrite computes the same underlying values; the difference is display precision only.

4 Bugs in the original archive

4.1 Bug 1: `listsimulator` undefined in `ACCG_list_placebo_I_power.R`

The power simulation script calls `listsimulator()` but the source file defines `list_simulator()` (with an underscore). All four `parSapply` calls fail with `Error: could not find function "listsimulator"`.

```
# Bug (ACCG_list_placebo_I_power.R, lines 41, 83, 124, 166)
x <- listsimulator(N.trueadmitter = ...) $reject_null

# Fix
x <- list_simulator(N_trueadmitter = ...) [["reject_null"]]
```

4.2 Bug 2: Dot-separated argument names

The power script passes `N.trueadmitter`, `N.falseconfessor`, `N.withholder`, `N.innocent`, `pr.listliar`, and `pr.designaffected` — all using dots — but the function signature uses underscores throughout. In base R, these are silently ignored via partial matching in some contexts but fail when the names are unambiguous.

```
# Bug
list_simulator(N.trueadmitter = N.trueadmitter, pr.listliar = listliar_prop)

# Fix
list_simulator(N_trueadmitter = N_trueadmitter, pr_listliar = listliar_prop)
```

4.3 Bug 3: `$p_value` on an atomic vector in `ACCG_list_source.R`

Inside `list_simulator()`, the `placeboI_test()` result is accessed with `$p_value`:

```
# Bug (ACCG_list_source.R, line 112)
reject_null <- as.numeric(placeboI_test(V=V, Z=Z, Y=Y)$p_value < 0.05)
```

`placeboI_test()` returns a named numeric vector (from `return(c(est=est, ...))`), not a list. The `$` operator on a named vector returns `NULL` in R, so `reject_null` is always `NULL` rather than 0 or 1. Power estimates from the original script are therefore all `NA`. The fix is `[["p_value"]]`, which works on both named vectors and lists:

```r
# Fix
reject_null <- as.integer(placeboI_test(V=V, Z=Z, Y=Y)[["p_value"]] < 0.05)
```

This bug means the published Figure A1 cannot be reproduced from the original archive. The maintained rewrite corrects it; `placebo_i_test()` returns a tibble, and `[["p_value"]]` extracts the scalar correctly.

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5 Deprecated patterns replaced

Original	Replacement
`rm(list = ls())`	(omitted)
`setwd("/Users/Alex/...")`	`here::here()`
`with(df, ...)` for all computations	Explicit `df[["col"]]` references
`xtable()`	`tibble` + `write_csv` + console print
`lattice::levelplot`	`ggplot2::geom_tile`
`makeCluster` + `parSapply`	`parallel::mclapply`
`rbind(c(...), c(...), ...)` matrix accumulation	`map_dfr` + `whole_shebang()` returning a tibble
Dot-separated argument names	Underscore argument names throughout

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6 Figures

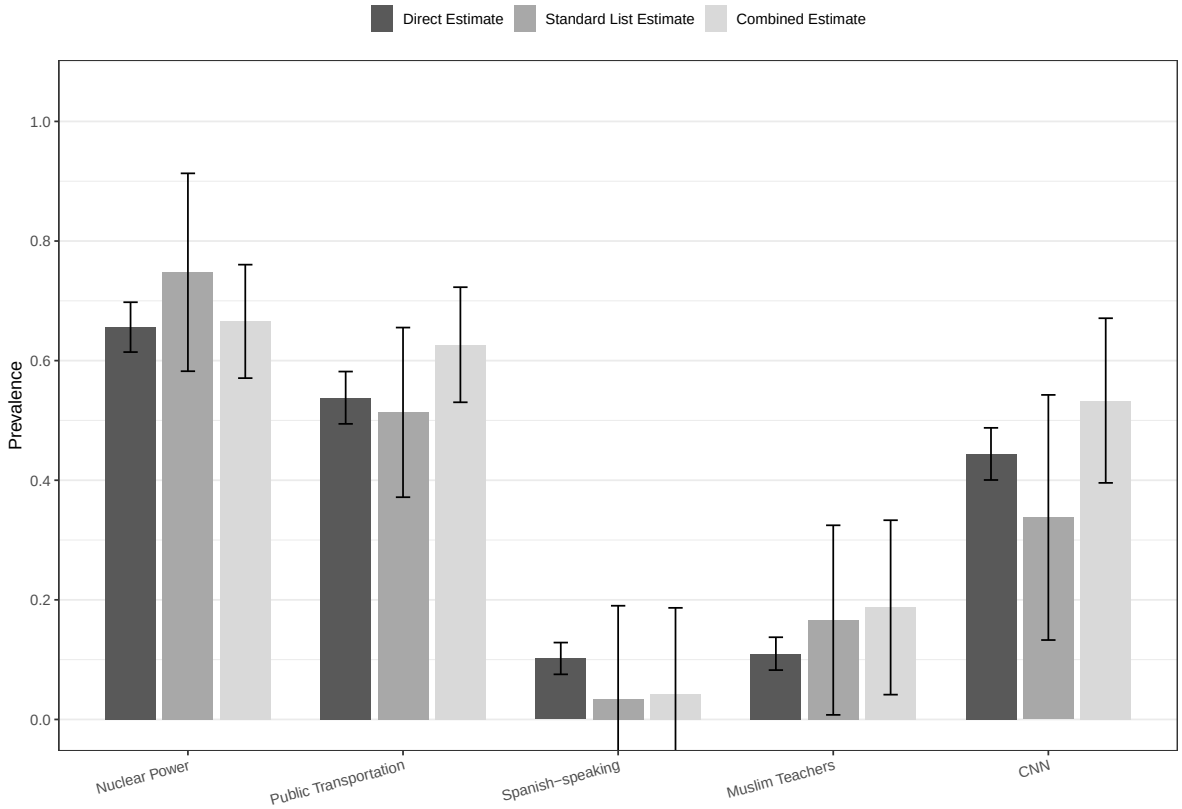


Figure 1: Figure 1 — Study A (Directs First): three estimates of prevalence with 95% CIs.

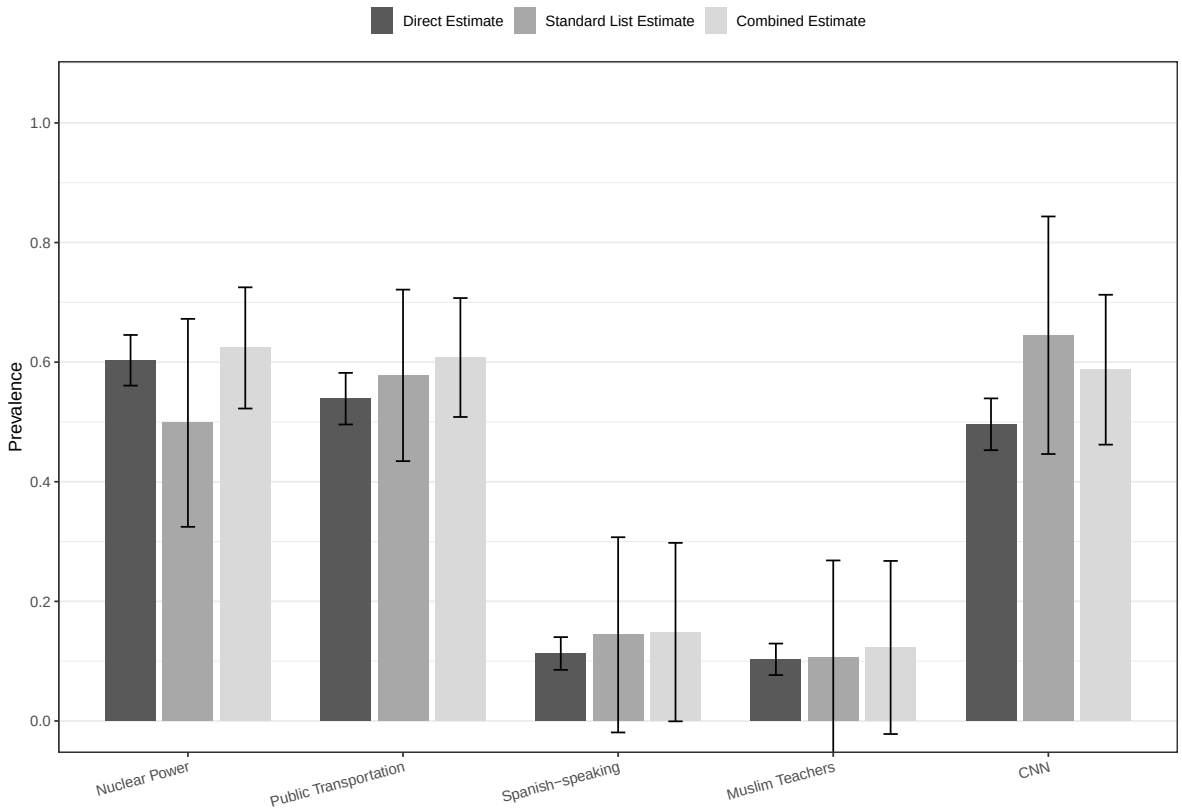


Figure 2: Figure 2 — Study B (Lists First): three estimates of prevalence with 95% CIs.

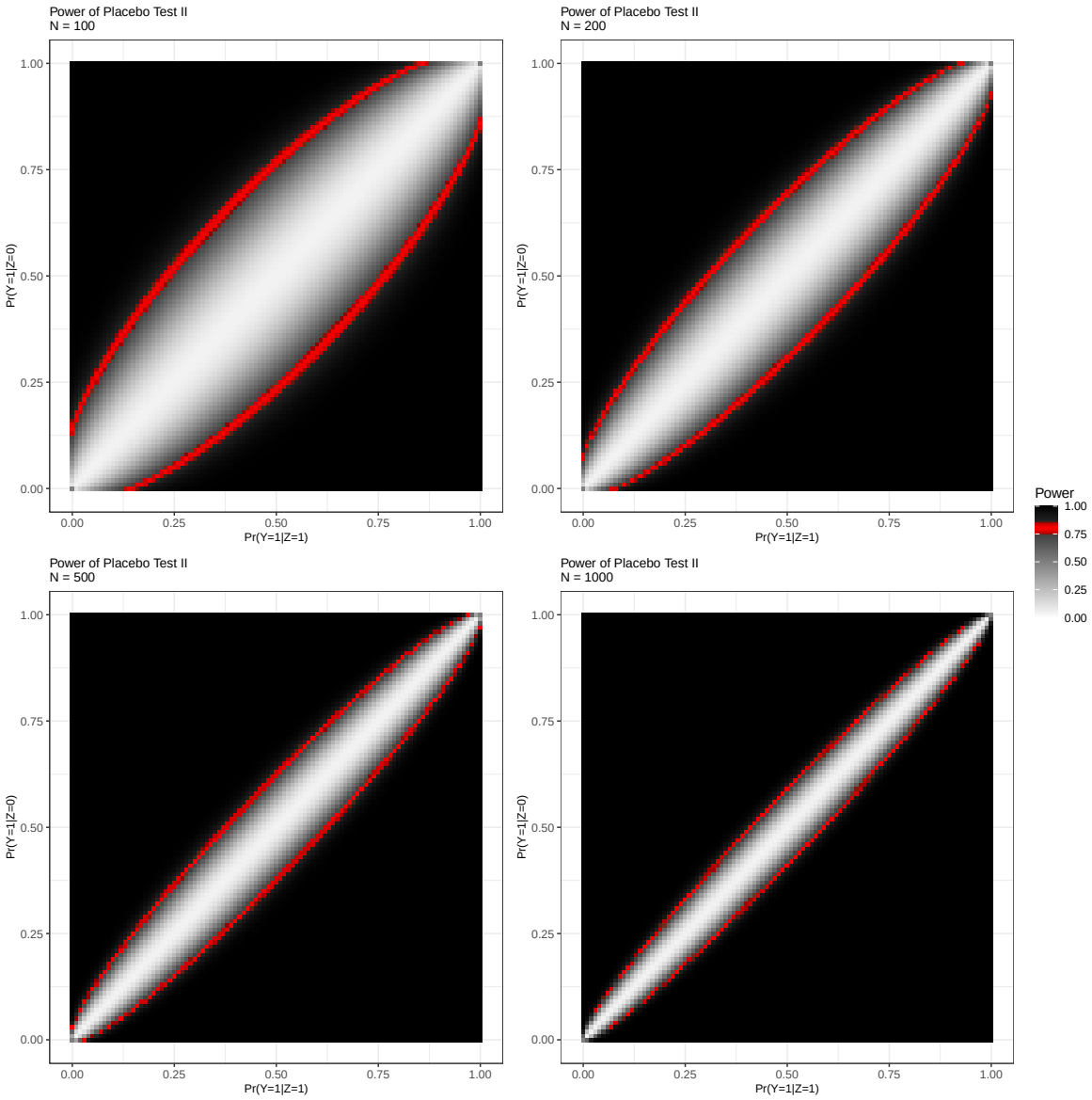


Figure 3: Figure A2 — Power of Placebo Test II to detect violations of Treatment Independence (analytical, four sample sizes).

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7 Selected tables

Table 4: Table 3: Study A (Directs First) — Three Estimates of Prevalence. $n = 500$.

Item	Direct	SE	Conv.	SE	Combined	SE	\% Var. Red.
Nuclear Power	0.656	0.021	0.748	0.084	0.666	0.048	67.1
Public Transportation	0.538	0.022	0.513	0.072	0.627	0.049	54.0
Spanish-speaking	0.102	0.014	0.035	0.079	0.042	0.074	14.0
Muslim Teachers	0.110	0.014	0.166	0.081	0.187	0.074	15.3
CNN	0.444	0.022	0.338	0.105	0.533	0.070	54.9

Table 5: Table 6: Study B (Lists First) — Three Estimates of Prevalence. $n = 514$.

Item	Direct	SE	Conv.	SE	Combined	SE	\% Var. Red.
Nuclear Power	0.603	0.022	0.499	0.089	0.624	0.052	66.0
Public Transportation	0.539	0.022	0.578	0.073	0.608	0.051	51.9
Spanish-speaking	0.113	0.014	0.144	0.083	0.149	0.076	16.4
Muslim Teachers	0.103	0.013	0.107	0.083	0.123	0.074	20.1
CNN	0.496	0.022	0.645	0.101	0.587	0.064	60.2

Table 6: Table 4: Study A Placebo Test I. H_0: $\beta = 1$ among direct admitters.

Item	$\hat{\beta}$	SE	p-value	n
Nuclear Power	1.054	0.095	0.568	328
Public Transportation	0.790	0.091	0.021	269
Spanish-speaking	0.848	0.279	0.585	51
Muslim Teachers	1.008	0.237	0.973	55
CNN	0.696	0.143	0.034	222

Table 7: Table 8: Study B Placebo Test II. H_0: $\delta = 0$ (treatment independence).

Item	$\hat{\beta}$	SE	p-value	n
Nuclear Power	-0.086	0.043	0.045	514
Public Transportation	0.034	0.044	0.434	514
Spanish-speaking	0.027	0.028	0.337	514
Muslim Teachers	0.016	0.027	0.549	514
CNN	0.132	0.044	0.003	514

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8 Maintained rewrite verification

Table 8: Maintained rewrite verification: paper value vs. rewrite output

Location	Quantity	Paper	Rewrite	Match
table_3	Study A Nuclear power direct est	0.656	0.6560000	1
table_3	Study A Nuclear power direct SE	0.021	0.0212658	1
table_3	Study A Nuclear power conv est	0.748	0.7477391	1
table_3	Study A Nuclear power conv SE	0.084	0.0844101	1
table_3	Study A Nuclear power combined est	0.666	0.6656184	1
table_3	Study A Nuclear power combined SE	0.048	0.0484034	1
table_3	Study A Nuclear power var reduction	67.100	67.1176027	1
table_3	Study A Public transportation direct est	0.538	0.5380000	1
table_3	Study A Public transportation direct SE	0.022	0.0223183	1
table_3	Study A Public transportation conv est	0.513	0.5134710	1
table_3	Study A Public transportation conv SE	0.072	0.0723675	1
table_3	Study A Public transportation combined est	0.627	0.6265818	1
table_3	Study A Public transportation combined SE	0.049	0.0490987	1
table_3	Study A Public transportation var reduction	54.000	53.9686591	1
table_3	Study A Spanish speaking direct est	0.102	0.1020000	1
table_3	Study A Spanish speaking direct SE	0.014	0.0135484	1
table_3	Study A Spanish speaking conv est	0.035	0.0347475	1
table_3	Study A Spanish speaking conv SE	0.079	0.0792860	1
table_3	Study A Spanish speaking combined est	0.042	0.0424747	1
table_3	Study A Spanish speaking combined SE	0.074	0.0735388	1
table_3	Study A Spanish speaking var reduction	14.000	13.9719777	1
table_3	Study A Muslim teachers direct est	0.110	0.1100000	1
table_3	Study A Muslim teachers direct SE	0.014	0.0140069	1
table_3	Study A Muslim teachers conv est	0.166	0.1661463	1
table_3	Study A Muslim teachers conv SE	0.081	0.0808654	1
table_3	Study A Muslim teachers combined est	0.187	0.1873131	1
table_3	Study A Muslim teachers combined SE	0.074	0.0744040	1
table_3	Study A Muslim teachers var reduction	15.300	15.3421496	1
table_3	Study A CNN direct est	0.444	0.4440000	1
table_3	Study A CNN direct SE	0.022	0.0222422	1
table_3	Study A CNN conv est	0.338	0.3377509	1
table_3	Study A CNN conv SE	0.105	0.1045596	1
table_3	Study A CNN combined est	0.533	0.5332502	1
table_3	Study A CNN combined SE	0.070	0.0702377	1
table_3	Study A CNN var reduction	54.900	54.8754050	1
table_4	Study A Placebo I Nuclear power beta	1.054	1.0539615	1
table_4	Study A Placebo I Nuclear power SE	0.095	0.0945273	1
table_4	Study A Placebo I Nuclear power p-value	0.568	0.5680971	1
table_4	Study A Placebo I Nuclear power n	328.000	328.0000000	1
table_4	Study A Placebo I Public transportation beta	0.790	0.7901152	1
table_4	Study A Placebo I Public transportation SE	0.091	0.0909323	1
table_4	Study A Placebo I Public transportation p-value	0.021	0.0209912	1
table_4	Study A Placebo I Public transportation n	269.000	269.0000000	1
table_4	Study A Placebo I Spanish speaking beta	0.848	0.8476190	1
table_4	Study A Placebo I Spanish speaking SE	0.279	0.2788110	1
table_4	Study A Placebo I Spanish speaking p-value	0.585	0.5846958	1
table_4	Study A Placebo I Spanish speaking n	51.000	51.0000000	1
table_4	Study A Placebo I Muslim teachers beta	1.008	1.0081522	1
table_4	Study A Placebo I Muslim teachers SE	0.237	0.2374838	1
table_4	Study A Placebo I Muslim teachers p-value	0.973	0.9726162	1
table_4	Study A Placebo I Muslim teachers n	55.000	55.0000000	1
table_4	Study A Placebo I CNN beta	0.696	0.6964286	1
table_4	Study A Placebo I CNN SE	0.143	0.1428606	1
table_4	Study A Placebo I CNN p-value	0.034	0.0335910	1

table_4	Study A Placebo I CNN n	222.000	222.0000000	1
table_5	Study A Placebo II Nuclear power beta	0.066	0.0662665	1
table_5	Study A Placebo II Nuclear power SE	0.042	0.0424397	1
table_5	Study A Placebo II Nuclear power p-value	0.118	0.1184234	1
table_5	Study A Placebo II Public transportation beta	0.000	-0.0003528	1
table_5	Study A Placebo II Public transportation SE	0.045	0.0447336	1
table_5	Study A Placebo II Public transportation p-value	0.994	0.9937072	1
table_5	Study A Placebo II CNN beta	-0.056	-0.0563562	1
table_5	Study A Placebo II CNN SE	0.045	0.0445317	1
table_5	Study A Placebo II CNN p-value	0.206	0.2056818	1
table_6	Study B Nuclear power direct est	0.603	0.6031128	1
table_6	Study B Nuclear power direct SE	0.022	0.0216010	1
table_6	Study B Nuclear power conv est	0.499	0.4985112	1
table_6	Study B Nuclear power conv SE	0.089	0.0887090	1
table_6	Study B Nuclear power combined est	0.624	0.6237910	1
table_6	Study B Nuclear power combined SE	0.052	0.0516956	1
table_6	Study B Nuclear power var reduction	66.000	66.0396534	1
table_6	Study B Public transportation direct est	0.539	0.5389105	1
table_6	Study B Public transportation direct SE	0.022	0.0220086	1
table_6	Study B Public transportation conv est	0.578	0.5778787	1
table_6	Study B Public transportation conv SE	0.073	0.0731664	1
table_6	Study B Public transportation combined est	0.608	0.6077617	1
table_6	Study B Public transportation combined SE	0.051	0.0507297	1
table_6	Study B Public transportation var reduction	51.900	51.9270084	1
table_6	Study B Spanish speaking direct est	0.113	0.1128405	1
table_6	Study B Spanish speaking direct SE	0.014	0.0139693	1
table_6	Study B Spanish speaking conv est	0.144	0.1440340	1
table_6	Study B Spanish speaking conv SE	0.083	0.0832810	1
table_6	Study B Spanish speaking combined est	0.149	0.1485933	1
table_6	Study B Spanish speaking combined SE	0.076	0.0761297	1
table_6	Study B Spanish speaking var reduction	16.400	16.4365866	1
table_6	Study B Muslim teachers direct est	0.103	0.1031128	1
table_6	Study B Muslim teachers direct SE	0.013	0.0134266	1
table_6	Study B Muslim teachers conv est	0.107	0.1065100	1
table_6	Study B Muslim teachers conv SE	0.083	0.0826027	1
table_6	Study B Muslim teachers combined est	0.123	0.1228877	1
table_6	Study B Muslim teachers combined SE	0.074	0.0738394	1
table_6	Study B Muslim teachers var reduction	20.100	20.0925211	1
table_6	Study B CNN direct est	0.496	0.4961089	1
table_6	Study B CNN direct SE	0.022	0.0220749	1
table_6	Study B CNN conv est	0.645	0.6450030	1
table_6	Study B CNN conv SE	0.101	0.1013401	1
table_6	Study B CNN combined est	0.587	0.5873763	1
table_6	Study B CNN combined SE	0.064	0.0639311	1
table_6	Study B CNN var reduction	60.200	60.2019370	1
table_7	Study B Placebo I Nuclear power beta	0.881	0.8813538	1
table_7	Study B Placebo I Nuclear power SE	0.113	0.1131145	1
table_7	Study B Placebo I Nuclear power p-value	0.294	0.2942226	1
table_7	Study B Placebo I Nuclear power n	310.000	310.0000000	1
table_7	Study B Placebo I Public transportation beta	0.913	0.9132728	1
table_7	Study B Placebo I Public transportation SE	0.091	0.0907433	1
table_7	Study B Placebo I Public transportation p-value	0.339	0.3392024	1
table_7	Study B Placebo I Public transportation n	277.000	277.0000000	1
table_7	Study B Placebo I Spanish speaking beta	0.767	0.7668269	1
table_7	Study B Placebo I Spanish speaking SE	0.229	0.2293168	1
table_7	Study B Placebo I Spanish speaking p-value	0.309	0.3092407	1
table_7	Study B Placebo I Spanish speaking n	58.000	58.0000000	1

table_7	Study B Placebo I Muslim teachers beta	0.700	0.7000000	1
table_7	Study B Placebo I Muslim teachers SE	0.285	0.2850436	1
table_7	Study B Placebo I Muslim teachers p-value	0.293	0.2925836	1
table_7	Study B Placebo I Muslim teachers n	53.000	53.0000000	1
table_7	Study B Placebo I CNN beta	0.847	0.8465502	1
table_7	Study B Placebo I CNN SE	0.135	0.1349657	1
table_7	Study B Placebo I CNN p-value	0.256	0.2555575	1
table_7	Study B Placebo I CNN n	255.000	255.0000000	1
table_8	Study B Placebo II Nuclear power beta	-0.086	-0.0862603	1
table_8	Study B Placebo II Nuclear power SE	0.043	0.0430536	1
table_8	Study B Placebo II Nuclear power p-value	0.045	0.0451177	1
table_8	Study B Placebo II Public transportation beta	0.034	0.0344159	1
table_8	Study B Placebo II Public transportation SE	0.044	0.0440371	1
table_8	Study B Placebo II Public transportation p-value	0.434	0.4344958	1
table_8	Study B Placebo II CNN beta	0.132	0.1324046	1
table_8	Study B Placebo II CNN SE	0.044	0.0438055	1
table_8	Study B Placebo II CNN p-value	0.003	0.0025065	1
table_a1	Factorial Nuclear power direct diff	0.053	0.0528872	1
table_a1	Factorial Nuclear power direct SE	0.030	0.0303123	1
table_a1	Factorial Nuclear power conv diff	0.249	0.2492279	1
table_a1	Factorial Nuclear power conv SE	0.122	0.1224514	1
table_a1	Factorial Nuclear power combined diff	0.042	0.0418274	1
table_a1	Factorial Nuclear power combined SE	0.071	0.0708190	1
table_a1	Factorial CNN direct diff	-0.052	-0.0521089	1
table_a1	Factorial CNN direct SE	0.031	0.0313372	1
table_a1	Factorial CNN conv diff	-0.307	-0.3072521	1
table_a1	Factorial CNN conv SE	0.146	0.1456109	1
table_a1	Factorial CNN combined diff	-0.054	-0.0541261	1
table_a1	Factorial CNN combined SE	0.095	0.0949765	1
table_a2	Replication Study A Nuclear power direct est	0.646	0.6462451	1
table_a2	Replication Study A Nuclear power conv est	0.575	0.5745938	1
table_a2	Replication Study A Nuclear power combined est	0.647	0.6471809	1
table_a2	Replication Study A Nuclear power var reduction	67.556	67.5563090	1
table_a2	Replication Study A CNN direct est	0.407	0.4071146	1
table_a2	Replication Study A CNN conv est	0.256	0.2556391	1
table_a2	Replication Study A CNN combined est	0.298	0.2984991	1
table_a2	Replication Study A CNN var reduction	45.350	45.3498613	1
table_a3	Replication Study A Placebo I CNN beta	0.711	0.7109848	1
table_a3	Replication Study A Placebo I CNN SE	0.133	0.1326367	1
table_a3	Replication Study A Placebo I CNN p-value	0.029	0.0293318	1
table_a3	Replication Study A Placebo I CNN n	206.000	206.0000000	1
table_a5	Replication Study B Nuclear power direct est	0.591	0.5909091	1
table_a5	Replication Study B Nuclear power conv est	0.364	0.3643623	1
table_a5	Replication Study B Nuclear power combined est	0.521	0.5206612	1
table_a5	Replication Study B Nuclear power var reduction	61.053	61.0531947	1
table_a5	Replication Study B CNN direct est	0.457	0.4565217	1
table_a5	Replication Study B CNN conv est	0.586	0.5858466	1
table_a5	Replication Study B CNN combined est	0.604	0.6044223	1
table_a5	Replication Study B CNN var reduction	51.321	51.3205255	1
table_a6	Replication Study B Placebo I Nuclear power beta	0.739	0.7388086	1
table_a6	Replication Study B Placebo I Nuclear power SE	0.110	0.1101300	1
table_a6	Replication Study B Placebo I Nuclear power p-value	0.018	0.0177081	1
table_a6	Replication Study B Placebo I Nuclear power n	299.000	299.0000000	1
table_a6	Replication Study B Placebo I Public transportation beta	0.795	0.7953967	1
table_a6	Replication Study B Placebo I Public transportation SE	0.097	0.0965730	1
table_a6	Replication Study B Placebo I Public transportation p-value	0.034	0.0341210	1
table_a6	Replication Study B Placebo I Public transportation n	263.000	263.0000000	1

table_a6	Replication Study B Placebo I Spanish speaking beta	0.301	0.3012821	1
table_a6	Replication Study B Placebo I Spanish speaking SE	0.262	0.2618676	1
table_a6	Replication Study B Placebo I Spanish speaking p-value	0.008	0.0076257	1
table_a6	Replication Study B Placebo I Spanish speaking n	50.000	50.0000000	1
table_a7	Replication Study B Placebo II CNN beta	0.092	0.0918846	1
table_a7	Replication Study B Placebo II CNN SE	0.044	0.0442387	1
table_a7	Replication Study B Placebo II CNN p-value	0.038	0.0377999	1
table_a8	Replication Factorial CNN conv diff	-0.330	-0.3302075	1
table_a8	Replication Factorial CNN conv SE	0.146	0.1458018	1
table_a8	Replication Factorial CNN combined diff	-0.306	-0.3059232	1
table_a8	Replication Factorial CNN combined SE	0.105	0.1045957	1

All **179** of **179** verifiable values produced by the maintained rewrite match the published paper to reported precision (`match_rewrite = 1`); 0 rows are unverifiable (`match_rewrite = NA`).

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9 R environment

Item	Value
R version	4.5.2
Platform	aarch64-apple-darwin20
Date run	2026-03-21